

Generation Interconnection Request System Impact Study Report

For Queue Project Q14-0370

Mickens 138 kV
September 25, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0370. The Interconnection Customer proposes a 556.5 MW natural gas combined cycle facility interconnecting at the Mickens 138 kV substation in the Virginia area. The request was received on August 28, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 232 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,714,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Mickens 345/138kV Transformer: 160.2% loading (565 MVA) following the outage of the Yadkin Substation to Mickens 138kV; DFAX = 73.1%.

The Yadkin to Harborgen Substation 138kV Line is overloaded to 139.8% of its 508 MVA emergency rating for the loss of the Wheelabrator Portsmouth to Yadkin 138kV. The project contributes 25.5% DFAX (141.9 MW) to this overload.

For the loss of Yadkin to Harborgen 138kV, the Yadkin to Wheelabrator Portsmouth 138kV Line loads to 134.4% of its 632 MVA rating. Project DFAX: 30.4%.

Trans Bay Cable Facility to Churchland 138kV L: 129.1% loading (382 MVA) following the outage of the Yadkin to Mickens 138kV; DFAX = 26.9%.

Benton City to Churchland 138kV: 123.3% loading (443 MVA) following the outage of the Yadkin to Mickens 138kV L; DFAX = 26.9%.

For the loss of Mickens Substation to Yadkin 138kV line, the Trans Bay Cable Facility Substation to Gosport Substation 138kV loads to 121.0% of its 311 MVA rating. Project DFAX: 15.0%.

Vly Arms to Benton City 138kV: 120.9% loading (475 MVA) following the outage of the Yadkin to Mickens 138kV; DFAX = 26.9%.

Beltsville to Bunker Hill 138kV: 120.5% loading (586 MVA) following the outage of the Taber to Glenannan 345kV; DFAX = 5.9%.

The High Ridge to Jerome 138kV line is overloaded to 120.2% of its 601 MVA emergency rating for the loss of the Taber Substation to Dixleta 345kV. The project contributes 8.1% DFAX (45.1 MW) to this overload.

For the loss of Taber to Dixleta 345kV, the N Pole to Dixleta 138kV Cir. 1 loads to 119.4% of its 338 MVA rating. Project DFAX: 5.1%.

For the loss of Yadkin to Mickens 138kV Line, the Valley Farms to Pt Roberts 138kV loads to 118.9% of its 504 MVA rating. Project DFAX: 26.9%.

For the loss of Glenannan Substation to Taber 345kV Line, the Nasa to Bunker Hill Substation 138kV L loads to 118.2% of its 629 MVA rating. Project DFAX: 5.9%.

Taber to Glenannan 345kV Line: 118.0% loading (1,656 MVA) following the outage of the Sierra Pacific Lumber 138/1 kV Transformer Bank; DFAX = 20.4%.

For the loss of Mickens to Swan 138kV line, the Yakin to Mickens 138kV Line loads to 117.6% of its 1,116 MVA rating. Project DFAX: 57.4%.

The Van Horne to Pt Roerts 138kV Line is overloaded to 117.5% of its 528 MVA emergency rating for the loss of the Mickens to Yadkin 138kV. The project contributes 26.9% DFAX (149.8 MW) to this overload.

Swan to Van Horne 138kV: 115.9% loading (558 MVA) following the outage of the Yadkin to Mickens 138kV; DFAX = 26.9%.

Bixby Hill to Sharyland 345kV line: 115.6% loading (1,851 MVA) following the outage of the Tombstone to Bixby Hill 500kV; DFAX = 7.3%.

Taber to Dixleta 345kV line: 115.5% loading (1,619 MVA) following the outage of the Sierra Pacific Lumber 138/1 kV Transformer; DFAX = 18.3%.

Swan to Mickens 138kV line: 114.8% loading (580 MVA) following the outage of the Yadkin to Mickens 138kV; DFAX = 26.9%.

Tombstone to Sharyland 345kV: 114.5% loading (1,905 MVA) following the outage of the Tombstone to Bixby Hill 500kV; DFAX = 7.2%.

Lake Creek to Sierra Pacific Lumber 138kV: 111.0% loading (612 MVA) following the outage of the Taber to Glenannan 345kV; DFAX = 5.5%.

For the loss of Taber to Glenannan 345kV Line, the Sunbright to Fairland Substation 138kV loads to 110.9% of its 673 MVA rating. Project DFAX: 7.1%.

The Sunbright to Colesville 138kV L is overloaded to 108.8% of its 736 MVA emergency rating for the loss of the Taber to Glenannan 345kV line. The project contributes 7.1% DFAX (39.8 MW) to this overload.

The Dixleta to Drum 138kV is overloaded to 107.2% of its 648 MVA emergency rating for the loss of the Glenannan to Taber 345kV. The project contributes 5.9% DFAX (33.0 MW) to this overload.

The Sierra Pacific Lumber 138/1kV TX is overloaded to 107.2% of its 963 MVA emergency rating for the loss of the Taber to Glenannan 345kV line. The project contributes 7.3% DFAX (40.5 MW) to this overload.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Mickens 345/138 kV Autotransformer	Replace transformer with higher-rated unit	\$21,886,000	35% (\$7,753,000)
Yadkin to Harborgen 138kV L	Rebuild line with higher-capacity conductor	\$3,763,000	30% (\$1,125,000)
Yadkin to Wheelabrator Portsmouth 138kV Line	Rebuild line with higher-capacity conductor	\$5,700,000	30% (\$1,694,000)
Trans Bay Cable Facility to Churchland 138kV	Rebuild line with higher-capacity conductor	\$692,000	35% (\$245,000)
Benton City to Churchland 138kV	Rebuild line with higher-capacity conductor	\$4,173,000	35% (\$1,478,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Trans Bay Cable Facility to Gosport 138kV	Rebuild line with higher-capacity conductor	\$2,943,000	28% (\$837,000)
Vly Arms to Benon City 138kV	Rebuild line with higher-capacity conductor	\$7,596,000	35% (\$2,690,000)
Beltsville to Bunker Hill 138kV	Rebuild line with higher-capacity conductor	\$868,000	10% (\$91,000)
Jerome to High Ridge 138kV	Rebuild line with higher-capacity conductor	\$2,756,000	11% (\$295,000)
N Pole to Dixleta 138kV Circuit 1	Rebuild line with higher-capacity conductor	\$1,140,000	12% (\$132,000)
Valley Farms to Point Roberts 138kV	Rebuild line with higher-capacity conductor	\$2,703,000	35% (\$957,000)
Bunker Hill to Nasa 138kV	Rebuild line with higher-capacity conductor	\$1,393,000	10% (\$146,000)
Taber Substation to Glenannan 345kV	Rebuild line with higher-capacity conductor	\$25,596,000	10% (\$2,587,000)
Yakin to Mickens 138kV Line	Rebuild line with higher-capacity conductor	\$3,203,000	33% (\$1,070,000)
Van Horne to Pt Roerts 138kV Line	Rebuild line with higher-capacity conductor	\$1,867,000	35% (\$661,000)
Swan to Van Horne 138kV	Rebuild line with higher-capacity conductor	\$2,808,000	35% (\$995,000)
Bixby Hill to Sharyland 345kV line	Rebuild line with higher-capacity conductor	\$30,026,000	9% (\$2,780,000)
Taber to Dixleta 345kV line	Rebuild line with higher-capacity conductor	\$22,369,000	10% (\$2,260,000)
Swan to Mickens 138kV line	Rebuild line with higher-capacity conductor	\$3,871,000	35% (\$1,371,000)
Tombstone to Sharyland 345kV	Rebuild line with higher-capacity conductor	\$47,687,000	9% (\$4,471,000)
Lake Creek to Sierra Pacific Lumber 138kV	Rebuild line with higher-capacity conductor	\$1,862,000	10% (\$195,000)
Sunbright to Fairland Substation 138kV	Rebuild line with higher-capacity conductor	\$406,000	11% (\$44,000)
Sunbright to Colesville 138kV L	Reconductor line	\$588,000	11% (\$63,000)
Dixleta to Drum 138kV	Reconductor line	\$838,000	11% (\$90,000)
Sierra Pacific Lumber 138 kV/1 kV Transformer Bank	Replace transformer with higher-rated unit	\$6,527,000	10% (\$684,000)
	Total allocated to project		\$34,714,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.